

# FINAL PROJECT: EVALUATING AND IMPROVING LLM SCIENTIFIC CAPABILITIES

CDS DS 595 · Spring 2026 · Boston University · [v0.1](#)

**35% of final grade**

## 1 Timeline

| Date       | Milestone                                                        |
|------------|------------------------------------------------------------------|
| Mon Mar 30 | <b>Project released.</b> Start (optional) team formation.        |
| Fri Apr 3  | <b>Teams finalized.</b> Students not in a team will be assigned. |
| Mon Apr 13 | <b>Proposal due.</b> Instructor feedback within 3 days.          |
| Apr 13–28  | Build eval, collect data, fine-tune, iterate.                    |
| Wed Apr 29 | <b>In-class presentations.</b>                                   |
| Fri May 1  | <b>Writeup due.</b>                                              |

## 2 Overview

In teams of 3, identify a **scientific capability** that current LLMs struggle with, build a **rigorous evaluation** for it, and **fine-tune an open-weights model** to improve that capability using the [Tinker](#) fine-tuning API.

The project has three phases:

1. **Propose** a capability and demonstrate that frontier models struggle with it
2. **Build** an evaluation benchmark and collect training data
3. **Fine-tune** a 4–8B parameter model and measure improvement

### Learning Objectives.

- Understand how scientific capabilities of LLMs are measured
- Design trustworthy evaluations with appropriate baselines
- Gain hands-on experience with LLM fine-tuning (SFT/RL)
- Get an understanding of the fine-tuning algorithms involved
- Develop intuition for token economics and compute budgets
- Learn to work effectively with LLMs as both tool and subject of study

## 3 What counts as a “scientific capability”?

Broadly interpreted! Almost any tasks which require domain knowledge and reasoning ability count. Some examples:

- Equivalence of complicated mathematical expressions – “is expression  $(a)$  the same as expression  $(b)$ ”?

- Aspects of reading a scientific plot/figure correctly (requires a multimodal model)
- Estimating the order of magnitude of physical quantities (hard Fermi problems)
- Computing scattering amplitudes from Feynman diagrams
- “Forecasting”, i.e. predicting future events given a knowledge of the past
- Saying “I don’t know how to do this” rather than giving a wrong answer. (Behaviour-grounded capabilities count too!)

You are encouraged to draw from your own scientific background and interests. **The main requirement is that the task should be something that frontier LLMs can’t already solve perfectly.** If you’re unsure whether your idea qualifies, ask!

## 4 Team Formation

Teams of 3 (teams of 2 considered on request). Use the Ed Discussion thread to find teammates. Teams must be finalized by **Friday, April 3**. Students without a team by that date will be assigned.

When teams are set, accept the GitHub Classroom assignment link (shared on Ed). The first team member creates the team name; others join.

## 5 Tinker

Each team receives **\$100 in API credits** for [Tinker](#), a distributed fine-tuning platform. Some recommended small models (feel free to pick your own; the model lineup is documented [here](#)):

- Qwen/Qwen3.5-4B
- meta-llama/Llama-3.1-8B

Think about how to allocate your compute credits: which model, how much training data, how many training runs. A 500-example SFT run on a 4B model costs roughly \$0.25, so you have plenty of room to iterate. Report your total spend and how it broke down in the writeup.

**Credits:** [TBD.]

**Getting started:** Sign up at <https://auth.thinkingmachines.ai/sign-up>. See the [Tinker Cookbook](#) for examples of supervised learning and RL workflows. If you use Claude Code, you can install the [Tinker skill](#) to get help navigating the API directly from your terminal.

## 6 Deliverables

### 6.1 Proposal (due Fri Apr 11)

A short document (PROPOSAL.md in your repo) containing:

1. **Capability:** One sentence describing the scientific task, what distinguishes it as an interesting and challenging task.
2. **Evidence of frontier failure:** ~ 5 concrete examples where a frontier model (Gemini, Claude, ...) gets the task wrong or performs poorly. Include the prompts and example outputs.
3. **Eval plan:** How will you measure success? What’s the input format, output format, and grading method?

4. **Data plan:** Where will your training data come from? Rough estimate of size. Do you plan to do SFT, RL?

## 6.2 Presentation (Wed Apr 29)

~ 8 minutes per team, followed by questions. Cover:

- What capability you targeted and why
- Your evaluation design
- Training approach and what you tried
- Results: frontier model vs. initial open-weights model vs. your fine-tuned model
- Conclusions, what worked and didn't

## 6.3 Writeup (due Fri May 1)

A GitHub repository containing all code, data, and a `README.md` writeup with the following sections:

1. **Capability.** What scientific task are you targeting? Why do LLMs struggle at it?
2. **Evaluation.** Description of your benchmark: how many examples, input/output format, how correctness is judged (exact match, rubric, automated checker). The eval script should be in the repo and runnable.
3. **Training.** What data did you use, how much, where did it come from? What model and method (SFT, RL)? How many tokens/steps? What did it cost?
4. **Results.** Three-way comparison on your eval:
  - A frontier model
  - The open-weights model before fine-tuning
  - Your fine-tuned model

Did fine-tuning help? On what types of problems? Any failure modes or regressions?

5. **Token economics.** Total Tinker spend and how it broke down (model choice, number of runs, tokens trained).
6. **Conclusions.** Reflections on what worked, what didn't, what you would do differently.
7. **AI usage.** Similar to previous assignments in the course, reflections on AI use.

## 7 Grading

**Total: 35 points (35% of final grade).** Eval design and fine-tuning scores will draw from both the writeup and the presentation.

| Component (points)             | Full marks                                                                                                                                                               | Needs improvement                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Proposal</b> (5 pts)        | Concrete capability with clear evidence that models fail; feasible data and eval plan                                                                                    | Capability is underspecified; few or no failure examples; data/eval plan needs more detail             |
| <b>Eval design</b> (10 pts)    | Well-designed benchmark (50+ examples, held out from training data); automated or clearly defined scoring; clear input/output format; appropriate baselines              | Eval is small or narrowly scoped; scoring criteria unclear; eval data not separated from training data |
| <b>Fine-tuning</b> (10 pts)    | Sound training methodology; meaningful iteration (tried multiple approaches, hyperparameters, data strategies); token spend reported; analysis of what worked and didn't | Limited iteration; few training runs attempted; token spend or analysis incomplete                     |
| <b>Presentation</b> (5 pts)    | Clear and well-structured; covers eval, training, and results; honest about what didn't work; handles questions                                                          | Some components unclear or missing; could better address questions about methodology                   |
| <b>Writeup</b> (2.5 pts)       | All sections present and clearly written; repo is organized and reproducible; thoughtful reflection                                                                      | Some sections incomplete; repo could be better organized; reflection is thin                           |
| <b>Peer feedback</b> (2.5 pts) | –                                                                                                                                                                        | Teammates flag unequal contribution                                                                    |

## 8 Peer Feedback

Each team member will have the opportunity to fill out a short form flagging significantly unequal contribution from teammates. If a form is not submitted or no issues are raised, everyone receives full marks. Scores may be additionally adjusted individually if there is a consistent pattern of unequal contribution.

## 9 AI Tools

You may use AI assistants freely throughout the project for code, data curation, writing, and analysis. In the README, briefly note how AI was used.